

Robust Predictions and Hard Information in the Market for Lemons

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Abstract

The literature on informationally-robust predictions focuses exclusively on “soft” information. In some environments, however, allowing for “hard” information is crucial. In a stylized setup of adverse selection, under certain conditions, we show that no-trade is the unique equilibrium outcome, however rich (soft) information structure we consider. With hard information, in contrast, there is an equilibrium where trade can occur, and the seller can obtain a positive expected payoff.

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1. Introduction

[Roesler and Szentes \(2017\)](#) and [Bergemann, Brooks and Morris \(2015\)](#) study stylized trading environments between a monopoly seller and buyer(s), interested in their *informationally-robust* predictions: what trading outcomes could be supported as a (Bayesian) equilibrium outcome if any admissible information structure is considered. [Roesler and Szentes \(2017\)](#) show that the optimal information structure for the buyer is not (necessarily) the one that the buyer knows everything: it may be better for him not to know his precise valuation. [Bergemann et al. \(2015\)](#) exhaust the set of possible equilibrium payoffs, assuming that the buyers know their precise values, while the seller may have imprecise information about them. [Kartik and Zhong \(2024\)](#) further generalize the approach to the case of interdependent values. Crucially, those papers (implicitly) assume that the considered information is “soft”. That is, there is no way that the holder of information can

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“prove” that piece of information. Consequently, any possible outcome in their predictions must satisfy the information holders’ *incentive compatibility* constraints.

In practice, however, information often takes the form of *evidence*, which may be shown to a third party as proof of that information. For example, a car manufacturer obtains a third-party certification for the quality of its car. As is well understood in the literature on hard information/evidence, such information may be elicitable without satisfying the same sort of incentive compatibility as soft information.¹ Therefore, the set of possible trading outcomes could be different if we allow for soft *and* hard information.

In this paper, we show in a stylized setup of adverse selection (Lemon) market à la [Akerlof \(1970\)](#) that such hard information is indeed critical in informationally-robust prediction. The buyer is assumed to have no information (as the seller in [Roesler and Szentes, 2017](#)). The seller may have any (possibly noisy) information about the quality. First, we consider an arbitrary “soft” information structure; that is, the seller may obtain an arbitrary signal about the quality of the good, but no evidence is attached to it. We show that, under certain conditions, no-trade is the unique equilibrium outcome, however rich (soft) information structure we consider. Next, we demonstrate by an example that, with hard information, non-trivial trade outcomes could be supported. Moreover, the difference is economically significant because the players’ payoffs and total surplus are strictly improving over the no-trade outcome. Based on this study, we conclude that it is important to allow for soft *and* hard information when we study informationally-robust predictions. In the adverse selection literature, the importance of hard information has been well-known (e.g., see [Viscusi, 1978](#)), though it often assumes some fixed information structure. Our point is that, even for informationally-robust prediction, it is crucial to explicitly consider soft and hard information (and not only for the stylized adverse selection environment but more generally). We hope this short note can be used as a small but first step toward a fuller understanding of this subject.

¹See [Green and Laffont \(1986\)](#) and [Bull and Watson \(2007\)](#).

2. Model

There is a seller (she) who has a good to sell to a buyer (he). For the good with quality (type) $q \in \{l, h\}$, the seller has valuation r_q , and the buyer has valuation v_q . Assume a common prior $\mu \in (0, 1)$ as the probability for $q = h$. If the good of type q is traded at price p , the seller gets $p - r_q$, and the buyer gets $v_q - p$; if no trade occurs, both get 0.

Assume that the valuations satisfy $v_l < r_l < r_h < v_h$; ex post efficiency requires trade to occur only for type h . Assume also that the prior μ satisfies

$$\mu < \bar{\mu} \equiv \frac{r_l - v_l}{v_h - r_h + r_l - v_l},$$

which implies that no trade occurs without any additional information. Indeed, if $\mu < \bar{\mu}$, the buyer's ex ante expected valuation $\mathbb{E}_\mu[v]$ is smaller than that of the seller, $\mathbb{E}_\mu[r]$:

$$\mathbb{E}_\mu[v] - \mathbb{E}_\mu[r] = \mu(v_h - r_h) + (1 - \mu)(v_l - r_l) < 0 \iff \mu < \bar{\mu}.$$

Thus, no mutually agreeable price can exist. Throughout the paper, we focus on this parameter range.

The seller may observe a signal about her valuation. An *information structure* is a pair (S, π) , where $S = \{s_1, s_2, \dots, s_n\}$ is a finite set of signals, and π is a joint distribution of type and signal. For each $q \in \{l, h\}$ and $i \in \{1, 2, \dots, n\}$, let π_{qi} denote the probability that signal s_i is realized and the good is of type q , where $\sum_{i=1}^n \pi_{qi} = 1$ for each $q \in \{l, h\}$. After the seller observes the realized signal, the buyer gives a take-it-or-leave-it price offer of p to the seller. A trade occurs if and only if the seller's expected valuation conditional on the realized signal exceeds p .²

3. Only soft information

Assume that the seller can obtain only soft information. Proposition 1 shows that if $\mu < \bar{\mu}$, equilibrium inefficiency is not amenable to the seller's learning of soft information about her type.

Proposition 1. *There exists no information structure under which trade occurs in equilibrium.*

²We take this seller's acceptance criterion for granted to simplify the analysis, but it is not crucial for the results.

PROOF. Consider any information structure (S, π) with n possible signals. For each $i \in \{1, 2, \dots, n\}$, let μ_i be the conditional probability for the high type given s_i :

$$\mu_i = \frac{\mu \pi_{hi}}{\mu \pi_{hi} + (1 - \mu) \pi_{li}}.$$

Order them so that, for $i \leq j$, $\mu_i \leq \mu_j$; equivalently,

$$\pi_{hi} \pi_{lj} \leq \pi_{hj} \pi_{li} \quad (1)$$

Since the seller with signal s_j accepts any offer $p \geq \mathbb{E}_{\mu_j}[r]$, the buyer's best option is to offer $p = \mathbb{E}_{\mu_j}[r]$ for some $j \in \{1, 2, 3, \dots, n\}$. The seller with any signal s_i with $i \leq j$ accepts it, yielding the buyer's expected payoff

$$\begin{aligned} & \mu \sum_{i=1}^j \pi_{hi} (v_h - \mathbb{E}_{\mu_j}[r]) + (1 - \mu) \sum_{i=1}^j \pi_{li} (v_l - \mathbb{E}_{\mu_j}[r]) \\ &= \left[\mu \sum_{i=1}^j \pi_{hi} + (1 - \mu) \sum_{i=1}^j \pi_{li} \right] (\mathbb{E}_{\mu_{\leq j}}[v] - \mathbb{E}_{\mu_j}[r]), \end{aligned}$$

where

$$\mu_{\leq j} = \frac{\mu \sum_{i=1}^j \pi_{hi}}{\mu \sum_{i=1}^j \pi_{hi} + (1 - \mu) \sum_{i=1}^j \pi_{li}}.$$

Note that (1) implies $\sum_{i=1}^j \pi_{hi} \pi_{lj} \leq \sum_{i=1}^j \pi_{hj} \pi_{li} \Leftrightarrow \mu_{\leq j} \leq \mu_j$.

- If $\mu_j < \mu < \bar{\mu}$, then

$$\mathbb{E}_{\mu_{\leq j}}[v] - \mathbb{E}_{\mu_j}[r] \leq \mathbb{E}_{\mu_j}[v - r] < 0.$$

- If $\mu_j \geq \mu$, then $\mu_i \geq \mu \Leftrightarrow \pi_{hi} \geq \pi_{li}$ for all $i \in \{j+1, \dots, n\}$, which implies $\sum_{i=j+1}^n \pi_{hi} \geq \sum_{i=j+1}^n \pi_{li}$. Since $\sum_{i=1}^n \pi_{qi} = 1$ for each $q \in \{l, h\}$, this in turn implies $\sum_{i=1}^j \pi_{hi} \leq \sum_{i=1}^j \pi_{li} \Leftrightarrow \mu_{\leq j} \leq \mu$. Thus,

$$\mathbb{E}_{\mu_{\leq j}}[v] - \mathbb{E}_{\mu_j}[r] \leq \mathbb{E}_{\mu}[v - r] < 0.$$

Therefore, no-trade is the unique equilibrium outcome, however rich information structure we consider.

4. Soft and hard information

Assume now that the seller can obtain and disclose hard information about her type. We demonstrate by an example that in equilibrium, trade can occur, and the seller can obtain a positive expected payoff. To do so, we consider the case with three possible signals $S = \{s_1, s_2, s_3\}$ such that $\mu_1 \leq \mu_2 \leq \mu_3$ and $\mu_1 \leq \mu \leq \mu_3$. The set of available evidence to the seller depends on the realized signal and is given as follows:

- If signal s_1 is realized, the seller has available $M_1 = \{x\}$;
- If signal s_2 or s_3 is realized, the seller has available $M_2 = M_3 = \{x, y\}$.

This evidence structure can be considered as a certification system; only when the seller receives signal s_2 or s_3 , she obtains certification y and can disclose it to the buyer. The timing is as follows:

1. Nature chooses the type of the good $q \in \{l, h\}$, but no player observes it;
2. Signal $s_i \in S$ is realized, and the seller observes it;
3. The seller discloses evidence $m \in M_i$ to the buyer;
4. The buyer offers a price p (accepted if and only if the seller's expected valuation given the realized signal exceeds it; otherwise, no trade).

The seller's strategy is a mapping $\alpha: S \rightarrow \{x, y\}$ such that $\alpha(s_i) \in M_i$ for each $s_i \in S$. The buyer's strategy is a mapping $\beta: \{x, y\} \rightarrow \mathbb{R}_+$. The seller's posterior belief for the high type is $\gamma: S \rightarrow [0, 1]$, where $\gamma(s_i) = \mu_i$ for each $i \in \{1, 2, 3\}$. The buyer's posterior belief about the realized type and signal is $\delta: \{x, y\} \rightarrow \Delta(\{l, h\} \times S)$. Given information structure (S, π) , profile $(\alpha, \beta, \gamma, \delta)$ of the players' strategies and belief system is a *weak perfect Bayesian equilibrium* (*weak PBE*) if

1. (α, β) is sequentially rational given (γ, δ) ; and
2. (γ, δ) is derived from (α, β) and (S, π) through Bayes' rule whenever possible.

Proposition 2 shows that the availability of hard information can amend equilibrium inefficiency.

Proposition 2. *There exists an information structure under which there is a weak PBE such that*

1. *trade occurs with positive probability; and*
2. *the seller obtains a positive expected payoff.*

PROOF. Consider signal distribution π^* represented by the following matrix:

$$\begin{pmatrix} 0 & \frac{r_l - v_l}{r_h - v_l} & \frac{r_h - r_l}{r_h - v_l} \\ \frac{r_h - v_l - \mu(v_h - v_l)}{(1 - \mu)(r_h - v_l)} & \frac{\mu(v_h - r_h)}{(1 - \mu)(r_h - v_l)} & 0 \end{pmatrix},$$

where $(1, i)$ entry is π_{hi}^* and $(2, i)$ entry is π_{li}^* . Observe $\mu_1 = 0$, $\mu_2 = \bar{\mu}$, and $\mu_3 = 1$. Consider the following strategies of the players:

- $\alpha^*(s_1) = x$ and $\alpha^*(s_2) = \alpha^*(s_3) = y$; and
- $\beta^*(x) = 0$ and $\beta^*(y) = \mathbb{E}_{\mu_3}[r] = r_h$.

There is a unique belief system (γ^*, δ^*) derived from (α^*, β^*) and (S, π^*) . Specifically, $\gamma^*(s_i) = \mu_i$ for each $i \in \{1, 2, 3\}$, $\delta^*(x)(l, s_1) = 1$, $\delta^*(y)(h, s_2) = \frac{r_l - v_l}{v_h - v_l}$, $\delta^*(y)(h, s_3) = \frac{r_h - r_l}{v_h - v_l}$, $\delta^*(y)(l, s_2) = \frac{v_h - r_h}{v_h - v_l}$, and $\delta^*(y)(q, s) = 0$ for any other $(q, s) \in \{l, h\} \times S$.

If the players follow (α^*, β^*) , trade occurs if signal s_2 or s_3 is realized, and the seller gets a positive expected payoff (she gets rent if signal s_2 is realized). Since the sequential rationality for the seller is clearly satisfied, we focus on the buyer's strategy β^* :

- If the buyer receives x , he believes any $p \geq \mathbb{E}_{\mu_1}[r] = r_l$ to be accepted. Hence, his expected payoff is $v_l - p \leq v_l - r_l < 0$. Thus, $\beta^*(x) = 0$ is optimal.
- If the buyer receives y , given δ^* , his best option is to offer either $\mathbb{E}_{\mu_2}[r] = \mathbb{E}_{\bar{\mu}}[r]$ or $\mathbb{E}_{\mu_3}[r] = r_h$. If he offers $\mathbb{E}_{\bar{\mu}}[r]$, only the seller with signal s_2 accepts it, yielding the buyer's expected payoff

$$\begin{aligned} & \frac{r_l - v_l}{v_h - v_l} (v_h - \mathbb{E}_{\bar{\mu}}[r]) + \frac{v_h - r_h}{v_h - v_l} (v_l - \mathbb{E}_{\bar{\mu}}[r]) \\ &= \frac{v_h - r_h + r_l - v_l}{v_h - v_l} \mathbb{E}_{\bar{\mu}}[v - r] = 0. \end{aligned}$$

If he offers r_h , the seller accepts it regardless of the realized signal, s_2 or s_3 , yielding the buyer's expected payoff

$$\frac{r_h - v_l}{v_h - v_l}v_h + \frac{v_h - r_h}{v_h - v_l}v_l - r_h = 0.$$

Thus, $\beta^*(y) = r_h$ is optimal.

Therefore, $(\alpha^*, \beta^*, \gamma^*, \delta^*)$ constitutes a weak PBE under (S, π^*) , where the seller captures the entire expected surplus.

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